

INTERVIEW PREPARATION

Technical Interview Prep Guide

C#, OOP fundamentals, application architecture, LINQ & collections, SQL databases, and professional development practices — 22 core questions answered.

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QUESTIONS

5

TOPIC AREAS

1 Object-Oriented Programming (C#)

1. What is the difference between a class and an object?

A A class is a blueprint — a category of things that share the same properties. An object is an actual instance built from that class, giving you access to the class's data.

2. What is encapsulation?

A Hiding private or internal data and restricting access to certain components, ensuring controlled interaction with objects.

3. Why should account balance not be public?

A If it's public, any part of the code can change it directly. Keeping it private forces the program to use safe, controlled methods instead.

4. What is the difference between a field and a property?

A A field is the basic variable used to store data. A property is a wrapper around a field that lets you add rules about who can read or change that data.

5. Why do we use constructors?

A Constructors are special methods that run exactly when an object is created, used to force an object to start with the required data.

2 Application Architecture

6. What is the purpose of a service class?

A A service class holds business logic and data management, separating heavy functionality and logic from the user interface.

7. Why should we avoid huge Main methods?

A Because it makes the code an unreadable nightmare to test and debug.

3 Collections & LINQ

8. What is the difference between List and Array?

A An array has a fixed size once created. A list is dynamic and can automatically grow or shrink as needed.

9. When would you use a Dictionary?

A When you have key-value pairs and need fast lookups.

10. What is LINQ used for?

A It's used to write short, clean, SQL-like queries directly in C#.

11. What is the difference between Where and Select?

A *Where* is used for filtering, while *Select* is used for projection / transformation.

12. What is GroupBy used for?

A It takes a list of data and organizes it into chunks based on a shared key.

13. What are Skip and Take used for?

A *Skip* ignores the first X results, *Take* grabs the next Y results.

4 Databases (SQL)

14. What is a primary key?

A A unique identifier for every single row in a database table.

15. What is a foreign key?

A A column in one table that links directly to the primary key of another table, creating a relationship between the two tables.

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16. What is a one-to-many relationship?

A One single record in Table A can be linked to multiple records in Table B.

17. Why do we use JOIN?

A Databases separate data into different tables to avoid duplication. We use JOIN to temporarily glue those tables back together in our queries.

18. What is the difference between a table and an entity?

A A "table" is the physical grid of rows and columns in the database. An "entity" is the corresponding C# class.

5 Professional Practices

19. Why do we use GitHub?

A To securely back up code, track the history of every change (version control), collaborate seamlessly with other developers, and act as a professional portfolio for employers.

20. What makes a README useful?

A A useful README acts as the "front door" to a project — clearly explaining what the project is, how to run it, and the technologies used.

21. Why are multiple commits better than one final commit?

A Multiple commits tell a clear, chronological story of how the app was built.

22. Why is professional delivery important?

A In the real world, writing code that works is only half the job — formatting code cleanly and organizing folders correctly matters just as much.